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MILITARY SPECIFICATION

RADIO SET AN/ARC-44( ) AND ASSOCIATED COMPONENTS: BENCH, PRE-FLIGHT AND FLIGHT TESTS OF

1. SCOPE

1.1 This specification covers methods and procedures for testing the performance and operation of Radio Set AN/ARC-44( ) and associated components both before and after installation in an aircraft.

1.2 List of Components for Test. - The following components shall meet the requirements specified herein. Whenever the type specification for a particular aircraft provides a list not in agreement with this list, the list in the aircraft type specification shall govern.

Receiver-Transmitter, Radio RT-294( )/ARC-44
Panel, Control SB-327( )/ARC-44
Dynamotor DY-107( )/AR
Mounting MT-1267( )/AR
Mounting MT-1268( )/U
Antenna AT-454( )/ARC
Test Set AN/ARM-68
Panel Signal Distribution, Radio SB-329( )/AR

2. APPLICABLE DOCUMENTS

2.1 There are no applicable documents.

3. REQUIREMENTS

3.1 Acceptance. - Any equipment which fails to meet the performance requirements specified herein shall be considered unacceptable for military use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of Test. - The testing of the equipment shall be classified as follows:

4.1.1 Bench Tests. - Bench tests shall be conducted at the initial installation point to determine that the equipment has not been damaged or the performance or opetation has not been affected in shipping and handling.

4.1.2 Pre-Flight Tests. - Pre-flight tests shall be conducted at the initial installation point, after bench tests have been made and after the equipment has been installed in the aircraft, prior to flight test. These tests are conducted to determine that the equipment is installed properly, to make a final check of aircraft wiring and to check on vibration and electrical interference.

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4.1.3 Flight Tests. - Flight tests shall be conducted as soon as possible after pre-flight tests on all aircraft to check performance and operation of the equipment while the aircraft is in flight.

4.2 Test Equipment Required. - The following test equipment shall be used in performing the tests required by this specification:

27.5 volt DC power supply, minimum capacity, continuous 10 amperes, 25 amperes surge.

Panel, Signal Distribution, Radio SB-329( )/AIC

Signal Generator SG-12( )/U, or equal

General Radio Audio Output Power Meter, Type 1840A or equal

Headset-Microphone H-101( )/U, or equal

Cable, (5 feet) Coaxial, RG-58( )/U with connectors

Test Set AN/ARM-68

Test Set AN/ARM-8

RF Watt Meter TS-2609/U

4.3 Bench Tests. -

4.3.1 General. - A test bench, cabled to accomodate the units to be tested and various units of test equipment, should be provided. The cabling between units shall be in accordance with a typical aircraft wiring diagram and shall be so arranged that the component parts are accessible. The power source should be connected into the test bench wiring and should be protected by suitable fuses or circuit breakers. Should Test Set AN/ARM-68 be used, the power source input should be fed into the test set.

4.3.2 Test Procedure. -

4.3.2.1 External Visual Inspection. - Inspect each major unit (receiver-transmitter, controls, and associated components) of the radio set externally for evidence of shipping damage. Manipulate the controls on the control unit to determine that they operate freely.

4.3.2.2 Internal Visual Inspection. -

4.3.2.2.1 Open the side doors on the receiver-transmitter, remove the dust cover from the control units and terminal box. The associated components shall be checked and inspected.

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4.3.2.2.2 Tubes, fuses, and plug-in stages shall be seated in proper sockets; shields and hold-downs shall be in place. Captive screws or snaps securing sub-assemblies shall be tight. Interconnecting coaxial cables between sub-assemblies shall be properly and securely connected. There shall be no dirt, metal scraps or packing materials on any part of the equipment.

4.3.2.2.3 Replace dust covers and close side doors.

(NOTE: The receiver-transmitter shall not be operated with the side doors open unless it is necessary to make internal adjustments).

4.3.3 Operating Tests. -

4.3.3.1 Connect all major units to the bench test cabling of AN/ARM-68. Interconnect Head Microphone H-101( )/U and Panel Signal Distribution Radio SB-329( )/AIC through the test bench wiring to the major units. Connect the r.f. wattmeter to the antenna jack using the 5 foot length of RG-58( )/U cable. The power supply voltage shall be maintained at 27.5 volts D.C. (± 2%) at the input terminals of the Dynamotor DY-107( ) and Receiver-Transmitter, Radio RT-294( )/ ARC-44. This input voltage shall be maintained throughout the operating tests.

4.3.3.2 Test Frequencies. - The equipment shall be tested for operation on the following frequencies:

Frequency - Mc	Frequency - Mc
24.0	39.7
26.1	41.8
28.2	43.9
30.3	45.0
32.4	47.1
34.5	49.2
36.6	51.3

4.3.3.3 Squelch Test. -

4.3.3.3.1 Note the position of the squelch adjust control. Rotate the squelch control on the front of the Receiver-Transmitter, Radio RT-294( )/ARC-44 counter-clockwise. A sudden hissing sound should be heard in Headset-Microphone H-101( )/U. Return the control to the above noted clockwise position.

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4.3.3.3.2 Squelch Adjustment. - Disconnect the signal generator. Connect the r.f. wattmeter to the antenna jack through a five foot length RG-58( )/U cable. The wattmeter with cable simulates the actual antenna load. Disconnect the audio output meter and connect Headset-Microphone H-101( )/U to Panel, Signal Distribution Radio SB-329( )/AIC. Set the controls as follows:

Control	Position
Power On-Off	On
Rem. Local	Local
Receiver Switch	#1 up, others down
Trans.	#1
Vol.(SB-327( )/ARC-44)	Maximum clockwise
Squelch	Maximum counter-clockwise

Adjust the squelch control in the clockwise direction to the point where the noise is just squelched off.

4.3.3.3.3 Squelch Sensitivity. - Disconnect wattmeter and connect the signal generator to the antenna jack. The Headset-Microphone H-101 ( )/U remains connected to the equipment. Set the controls as follows:

Control	Position
Power On-Off	On
Rem. Local	Local
Receiver Switch	#1 up, others down
Trans.	#1
Volume (SB-327( )/ARC-44)	Maximum clockwise
Squelch	Previously adjusted in 4.3.3.6

4.3.3.3.4 Adjust signal generator to 37 mc. Adjust signal generator to 0.6 microvolt. It shall take no more than 0.6 microvolt to unsquelch the receiver. The receiver shall be considered to be unsquelched when the audio output is within 2 db of the output obtained when the squelch control is rotated fully counterclockwise.

4.3.3.3.5 Microphone Test. - Depress the microphone switch and speak into the microphone. The operator's voice should be audible in the headset.

Caution: Do not use the microphone switch at any time without Test Set AN/ARM-68 or suitable Dummy Load at the transmitter output.

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4.3.4 Sensitivity. - The following frequencies are to be used to check the sensitivity:

Frequency-Mc	Frequency-Mc
24.0	39.7
26.1	41.8
28.2	43.9
30.3	45.0
32.4	47.1
34.5	49.2
36.6	51.3

4.3.4.1 Receiver-Audio Output and Sensitivity. - Connect the audio output meter to the 150 ohm audio output circuit of Receiver-Transmitter RT-294( )/ARC-44 and set the output meter for 150 ohm load impedance. Connect the signal generator (52 ohms impedance) to the antenna jack of the receiver-transmitter. Set the controls as follows:

Control	Position
On-Off	On
Rem. Local	Local
Receiver Switches	#1 up, all others down
Trans.	#1 position
Vol. (SB-327( )/ARC-44)	Maximum clockwise position
Squelch	Maximum counter clockwise

4.3.4.2 Set the signal generator attenuator to provide 100 microvolts and modulation controls to obtain 20 kc deviation frequency modulation at 1000 cycles per second at the resonant frequency on the receiver. Adjust the RF output frequency of the signal generator to the resonant channel frequency under test by adjusting the frequency dial of the signal generator until a maximum receiver audio output is obtained. The audio output should not be less than 50 milliwatts.

4.3.4.3 Set the receiver-transmitter to the channel to be checked. Set the signal generator attenuator to 0.6 microvolts and the modulation control to obtain 20 kc deviation frequency modulation at 1000 cycles per second. Adjust the frequency of the signal generator to the resonant frequency of the receiver-transmitter. Note the reading of the output meter. Now turn off the modulation of the signal generator, but maintain the RF carrier. The receiver audio output with no modulation shall be not more than one tenth of the receiver audio power output with 20 kc deviation frequency modulation at 100 cycles per second.

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4.3.5 Homing Output and Sensitivity. - Connect the audio output meter to the 150 ohm audio output circuit of Receiver-Transmitter, Radio RT-294( )/ARC-44 and set the output meter for 150 ohm load impedance. Connect the signal generator (52 ohms impedance) to the antenna jack of Receiver-Transmitter, Radio RT-294( )/ ARC-44. Connect a switch between ground and contact "M" on the multiple connector located at the read of the receiver transmitter. This switch will be referred to as the "homing switch". Set the controls as follows:

Controls	Position
On-Off	On
Rem. Local	Local
Receiver Switches	#1 up, all others down
Trans.	#1
Vol. SB-327( )/ARC-44)	Maximum clockwise
Squelch	Maximum counter-clockwise
Homing	Maximum clockwise
Homing Switch	Closed

4.3.5.1 Set the receiver-transmitter to the channel to be checked with the homing switch in the open position. Set the Signal Generator attenuator to 1.0 microvolts. Adjust the frequency of the signal generator to the resonant frequency of the receiver-transmitter. Adjust volume control on SB-327 for 50 millivolts output. Close homing switch and remove modulation and adjust homing gain control on receiver-transmitter to obtain 50 milliwatts. Open homing switch but maintain the RF carrier. The receiver audio output without tone modulation shall be not more than one-tenth of the receiver audio power output with homing switch closed.

WARNING: Do not transmit while signal generator is connected.

4.3.6 Power Output. - Disconnect the signal generator and connect the r.f. wattmeter TS-2609/U to the receiver-transmitter using the 5 foot length of RG-58( )/ U. Press the push-to-talk control attached to Headset-Microphone H-101( )/U. Note the power output reading indicated on the 52 ohm wattmeter for all test frequencies listed in 4.3.3.2. The power shall not be less than 8 watts when measured at normal room temperature of plus 70° F and not less than 4 watts when measured at extreme temperature of minus 40° F and plus 132° F.

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4.4 Pre-Flight Tests. -

4.4.1 General. - These tests shall be made with the radio set installed in accordance with the approved drawings and interconnecting diagrams, in the aircraft as an operating system.

4.4.2 Connect the external power source to the aircraft. The source voltage shall be adjusted to supply approximately 27.5 volts d.c. to the radio set.

4.4.3 Set the controls to the following positions:

Control	Position
Power On-Off	On
Rem. Local	Local
Receiver Switches	#1 up, all others down
Trans. Switch	#1 position
Vol. (SB-327( )/ ARC-44 or SB-329( ) /AIC)	Comfortable audio level
Frequency Selector	Authorized frequency for test

4.4.4 Use Headset-Microphone H-101( )/U adjust the boom so that the microprophone just touches the lips of the operator. Press the Push-to-Talk switch attached to the headset-microphone and speak clearly into the microphone. The operator's voice shall be audible in the headset. Repeat on all channels to be authorized for flight test.

4.4.5 Turn off the equipment and disconnect the external power source.

4.4.6 Before starting engine, make sure all safety precautions are taken. Start the aircraft engine(s) and operate at a speed sufficient to cause the generator(s) to charge the batteries. Turn the radio set "on". Turn on all other electronic equipment to be operated in the aircraft. Adjust the squelch control as outlined in 4.3.3.6 titled "Squelch Adjustment" except that the aircraft antenna replaces the r.f. wattmeter. Check on all channels authorized for use during the flight test.

4.4.7 Turn off the radio set, other electronic equipment and aircraft engine(s).

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4.5 Flight Tests. - Flight tests on all aircraft shall consist of normal operational use of the radio set during flight tests of the aircraft.

5. PREPARATION FOR DELIVERY

5.1 Not applicable.

6. NOTES

6.1 Principal Use. - Radio Set AN/ARC-44( ) is a remote-controlled, frequency-modulated, multichannel communciation set designed to operate over the frequency range of 24.0 to 51.9 megacycles. Circuits are included in the equipment to use the Homing Adapter AN/ARA-31( ) with this radio set for homing navigation. The radio set can communicate and net with Radio Sets AN/GRC-3( ) thru 8( ), Radio Sets AN/PRC-6( ), 8( ), 9( ), and 10( ).

6.2 Scope of Servicing and Repairs. - No servicing and repairs are required. They are functions of USAECOM.

6.3 It is desirable that personnel making the tests described herein be thoroughly familiar with the operating characteristics of Radio Set AN/ARC-44( ).

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Preparing Activity
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